smartwilds-bioacoustics — dataset card (review draft)

Rendered from
release/hf_smartwilds_20261022/smartwilds-bioacoustics/README.md

Dataset Card for SmartWilds Bioacoustics



A Wildlife Acoustics Song Meter Micro 2 (SM03) deployed in a tree at The Wilds.

Every retained acoustic recording from 10 recorder deployments at The Wilds in summer 2025, the schedule each recorder actually ran, and the birds BirdNET heard in each file.

This is the bioacoustics dataset of the **SmartWilds** collection. Each modality has its own dataset and DOI. They share one Darwin Core Event hierarchy rooted at `TheWilds_2025` and are joined by the SmartWilds collection **[TODO: confirm collection URL]**.

Dataset Details

- **Curated by:** Jenna M. Kline, Anirudh Potlapally, Elizabeth G. Campolongo
- **Homepage:** https://github.com/Imageomics/wilds_multimodal_data
- **Paper:** SmartWilds: A Systems Dataset and Metadata Foundation for Cross-Modal Ecological Sensing (in preparation)
- **Snapshot:** `darwin_core_latest_20261008`, release build 2026-10-22

Campaign	Dates	Deployments	WAV released
1	2025-06-30 to 2025-07-03	4	311

Campaign	Dates	Deployments	WAV released
2	2025-08-04 to 2025-08-28	6	2,060

Darwin Core tables. The Event table has 2,382 rows: 1 `Project`, 10 `SensorDeployment`, and 2,371 `AudioRecording` Events, one per WAV file with its start time. There are 9,960 Occurrences. Each Occurrence is one **recording × species** pair, and `individualCount` gives the number of 3-second BirdNET detections of that species in the file (93,509 in total).

Supported Tasks

- **Bird audio classification and re-labelling:** the raw WAV files plus the BirdNET settings let you rerun BirdNET or swap in another classifier.
- **Duty-cycle and acquisition-workload studies:** each recorder ran a different schedule (5 min/h, 15 min/h, sunrise-sunset continuous, 1 h at dawn and dusk), so `metadata/deployments.csv` supports counterfactual sampling-effort analyses.
- **Cross-modal comparison:** recorders sit beside the camera traps at TW02, TW06, and TW09.

Dataset Structure

```
smartwilds-bioacoustics/
├── data/
│   ├── events.csv          # Project, SensorDeployment, AudioRecording Events
│   ├── occurrences.csv     # one row per recording x species
│   └── recordings.csv      # one row per WAV path on disk (2,453): disposition
├── media/                  # media/<campaign>/<deployment>/<sd-card>/*.wav
├── media_manifest.csv
├── metadata/
│   ├── deployments.csv     # recorder, site, GPS, schedule, WAV counts
│   ├── birdnet_settings.csv
│   └── provenance/         # raw-WAV reconciliation report, repair audit
└── scripts/
```

`associatedMedia` on each `AudioRecording` Event and on each Occurrence gives the release-relative path of the WAV file.

Sensor settings

All recorders are Wildlife Acoustics **Song Meter Micro 2**, 48 kHz. `metadata/deployments.csv` records the schedule each recorder actually ran. TW05-SM01 in Campaign 1 was configured for 1 h at sunrise and sunset but recorded 5-minute clips every hour. The release documents the actual schedule and notes the intended one.

Schedule	Duty cycle	Recorders
5 min every hour	8.3%	TW05-SM01, TW06-SM03, TW08-SM04 (C1); TW02-SM04 (C2)
15 min every hour	25%	TW06-SM05 (C2)
Sunrise to sunset, continuous	~50-60%	TW05-SM06, TW07-SM02, TW08-SM03, TW09-SM01 (C2)
1 h at sunrise and sunset	~8%	TW07-SM02 (C1)

Annotation method

- **Method:** automated. No human verification (all rows are `unverified`).

- **Classifier:** BirdNET Analyzer v2.4, with 3.0 s segments, a 48 kHz sample rate, and a 0-15 kHz bandpass.
- **Taxon fields:** filled from BirdNET common-name mappings.
- **Coordinates:** the recorder location, not the animal location.

Dataset Creation

Raw-WAV reconciliation

There are 2,453 WAV paths on disk. `data/recordings.csv` classifies every one:

Disposition	Files
Released, with detections	2,307
Released, BirdNET ran but detected nothing (Event, no Occurrence)	64
Excluded: Campaign-1 files copied into a Campaign-2 directory (transfer duplicates)	78
Excluded: no BirdNET result, either truncated (3.9-22 s) or unreadable	4

The zero-detection recordings are kept as Events. This lets a user tell "recorded, nothing heard" apart from "not recorded".

Considerations for Using the Data

- **`individualCount` counts detections, not birds.** A singing bird produces many 3-second detections.
- **No confidence threshold is recorded per Occurrence.** Re-run BirdNET from the WAV files if you need score-level control.
- **Acquisition is driven by the schedule.** Arrival burstiness is low ($B \approx +0.10$). Detection burstiness is high ($B \approx +0.53$) and concentrated in the dawn chorus.

Supplemental Material

Extended inventories, the timestamp-repair methods, the standards crosswalk, validation checks, and the full defect inventory are in `SUPPLEMENTAL.md` in the study-level SmartWilds dataset. **[TODO: confirm repo URL]**

Licensing

Data: CC0 1.0 **[TODO(license): confirm across the collection]**. Scripts: MIT.

Citation

```
@misc{smartwilds2025_bioacoustics,
  author    = {Kline, Jenna M. and Potlapally, Anirudh and Campolongo, Elizabeth G.},
  title     = {SmartWilds Bioacoustics (The Wilds 2025)},
  year      = {2026},
  publisher = {Hugging Face},
  doi       = {TODO},
  url       = {https://huggingface.co/datasets/imageomics/smartwilds-bioacoustics}
}
```

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Dataset Card Contact

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Appendix: Hugging Face metadata (YAML front matter)

```yaml
license: cc0-1.0
language:
- en
pretty_name: SmartWilds Bioacoustics (The Wilds 2025)
task_categories:
- audio-classification
tags:
- biology
- ecology
- biodiversity
- bioacoustics
- passive-acoustic-monitoring
- birds
- birdnet
- darwin-core
- humboldt-eco
- fair2
- smartwilds
- the-wilds
- ohio
size_categories:
- 1K<n<10K
description: >-
  2,371 WAV recordings (215 GB) from 10 Song Meter Micro 2 deployments at six
  sites at The Wilds (Ohio, USA), June-August 2025, with per-recorder duty
  cycles, a one-row-per-file reconciliation table, and 9,960 Darwin Core
  Occurrences (93,509 BirdNET detections across 128 bird species).

fair2_compliance:
  findable:
    doi: ""  # TODO: assign on publication
    metadata_registry: ["DataCite", "GBIF"]
  accessible:
    open_access: true
    authentication_required: false
  interoperable:
    standards: ["Darwin Core", "Humboldt Eco", "FAIR2"]
  reusable:
    license_clear: true
    provenance_documented: true
  ai_ready:
    machine_readable: true
    structured_annotations: true
```

```yaml
darwin_core:
  event_coverage:
    start_date: "2025-06-30"
    end_date: "2025-08-28"
    decimal_latitude: 39.83
    decimal_longitude: -81.74
    coordinate_uncertainty_meters: 5000
    locality: "The Wilds, Cumberland, Muskingum County, Ohio, USA"
  occurrence_info:
    basis_of_record: "MachineObservation"
    taxa_included: "Aves (128 species)"
    sampling_protocol: "Scheduled passive acoustic recording; BirdNET Analyzer v2.4 classification"

sensors:
  - type: "Autonomous recording unit"
    manufacturer: "Wildlife Acoustics"
    model: "Song Meter Micro 2"

configs:
- config_name: occurrences
  data_files: data/occurrences.csv
- config_name: events
  data_files: data/events.csv
- config_name: recordings
  data_files: data/recordings.csv
```